

MA1508E Notes/Cheat Sheet

Brians Tjipto

LIST

Let A be an $n \times n$ matrix. The following are equivalent:

- A is invertible
- $\text{RREF}(A) = I_n$
- A is a product of elementary matrices.
- The system $A\mathbf{x} = \mathbf{0}$ has only the trivial solution ($\mathbf{x} = \mathbf{0}$). Equivalently, $\text{Null}(A) = \{\mathbf{0}\}$.
- The system $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $\mathbf{b} \in \mathbb{R}^n$.
- $\det(A) \neq 0$.
- The columns of A form a basis for $\mathbb{R}^n$ (and are thus Linearly Independent, L.I.).
- The rows of A form a basis for $\mathbb{R}^n$ (and are thus Linearly Independent, L.I.).
- 0 is not an eigenvalue of A .

MATLAB

```
format rat      % Display fractions
A = [1 0; 5 2; 0 0] % Define matrix A
A(2,:)         % Select second row
RREF(A)        % Reduced Row Echelon Form
inv(A)         % Inverse of A (if square & invertible)
A'             % Transpose of A
det(A)         % Determinant of A
syms x         % Define symbolic variable x
charpoly(A,x)  % Characteristic polynomial p(x)=det(xI-A)
eig(A)         % Eigenvalues and Eigenvectors
null(A)        % Basis for the null space (kernel)
```

1 LINEAR SYSTEMS

SYSTEM $\rightarrow$ MATRIX

Linear System in Standard Form:

$$\begin{aligned} 5x + y - 2z &= 7 \\ x + 7z &= 0 \end{aligned}$$

Augmented Matrix:
$$\left(\begin{array}{ccc|c} x & y & z & \text{const} \\ 5 & 1 & -2 & 7 \\ 1 & 0 & 7 & 0 \end{array} \right)$$

Associated Homogeneous Matrix: (RHS set to 0)

$$\left(\begin{array}{ccc|c} 5 & 1 & -2 & 0 \\ 1 & 0 & 7 & 0 \end{array} \right)$$

RREF (Reduced Row Echelon Form)

REF (Row Echelon Form) Example:

$$\left(\begin{array}{ccc|c} \boxed{1} & 0 & 5 & 7 \\ 0 & 0 & \boxed{2} & -1 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

Zero-Rows are at the bottom. As we go down, **leading entries** (first non-zero entry in a row, boxed) go right.

RREF Properties:

- It is in REF.
- All **leading entries** are 1.
- Each leading 1 is the only non-zero entry in its column (**pivot column**).

Example RREF: (From notes page 2)

$$\left(\begin{array}{cccc|c} 0 & \boxed{1} & 0 & 2 & 3 \\ 0 & 0 & \boxed{1} & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

* Matrices with the same RREF are **row equivalent**.

Inconsistent Linear System: Has no solution. This occurs if the RREF of the augmented matrix contains a row like $[0 \ 0 \ \dots \ 0 \ | \ b]$ where $b \neq 0$.

Example leading to inconsistency:
$$\left(\begin{array}{cc|c} 1 & 5 & 3 \\ 0 & 0 & 2 \end{array} \right)$$

Row implies $0x + 0y = 2$ (Impossible) **No Solution.**

GAUSSIAN ELIMINATION

Process of using Elementary Row Operations (EROs) to transform a matrix into REF or RREF. **Example:**

$$\begin{aligned} \left(\begin{array}{ccc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array} \right) &\xrightarrow{\substack{R_2 \leftarrow R_2 - 4R_1 \\ R_3 \leftarrow R_3 - 7R_1}} \left(\begin{array}{ccc} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -18 \end{array} \right) \xrightarrow{R_3 \leftarrow R_3 - 2R_2} \left(\begin{array}{ccc} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & -6 \end{array} \right) \\ &\xrightarrow{-\frac{1}{3}R_2} \left(\begin{array}{ccc} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & -6 \end{array} \right) \xrightarrow{-\frac{1}{6}R_3} \left(\begin{array}{ccc} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{array} \right) \\ &\xrightarrow{R_1 \leftarrow R_1 - 2R_2} \left(\begin{array}{ccc} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{array} \right) \xrightarrow{R_1 \leftarrow R_1 + R_3} \left(\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{array} \right) \xrightarrow{R_2 \leftarrow R_2 - 2R_3} \left(\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right) \end{aligned}$$

Solving Systems

First, bring the augmented matrix to RREF.

① No Solution

(As described before) Occurs if RREF has a row $[0 \ 0 \ \dots \ 0 \mid b]$ with $b \neq 0$.

e.g., $x + y = 0, 2x + 2y = 5 \implies \left(\begin{array}{cc|c} 1 & 1 & 0 \\ 2 & 2 & 5 \end{array} \right) \rightarrow \left(\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 0 & 5 \end{array} \right) \implies \text{No Solution}$

② Unique Solution

Exactly one pivot column for each variable (no free variables).

e.g., RREF:
$$\left(\begin{array}{ccc|c} x & y & z & \\ 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & -2 \end{array} \right) \implies \begin{cases} x = 3 \\ y = 5 \\ z = -2 \end{cases}$$

⊗ Infinite Solutions

Occurs when there are **non-pivot columns** (corresponding to variables) before the augmentation line. These columns correspond to **free variables**. Set **parameters** (e.g., $s, t \in \mathbb{R}$) for free variables to express the **general solution**.

e.g., RREF:
$$\left(\begin{array}{ccc|c} x & y & z & \\ 1 & 2 & 0 & -3 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

Here, y is a free variable. Let $y = s$, where $s \in \mathbb{R}$. Row 1: $x + 2y = -3 \implies x = -3 - 2y = -3 - 2s$. Row 2: $z = 4$. General Solution:
$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -3 - 2s \\ s \\ 4 \end{pmatrix} = \begin{pmatrix} -3 \\ 0 \\ 4 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}. \text{ (Bring terms involving parameters over from the left side).}$$

ELEMENTARY ROW OPERATIONS (EROs)

1. Swap two rows: $R_i \leftrightarrow R_j$.

$$\left(\begin{array}{cc} 0 & 1 \\ 1 & 0 \\ 0 & 0 \end{array} \right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{cc} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{array} \right)$$

2. Add a multiple of one row to another: $R_i \leftarrow R_i + kR_j$. (Multiplier k must be a real number).

$$I = \left(\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right) \xrightarrow{R_2 \leftarrow R_2 + 4R_3} \left(\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{array} \right)$$

3. Multiply a row by a non-zero scalar: $R_i \leftarrow cR_i$ ($c \neq 0$).

$$\left(\begin{array}{ccc} -3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right) \xrightarrow{R_1 \leftarrow (-1/3)R_1} \left(\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right)$$

2 MATRICES

BASIC TYPES

For an $m \times n$ matrix (m rows, n columns):

- **Diagonal:** $A_{ij} = 0$ if $i \neq j$.
- **Upper Triangular:** $A_{ij} = 0$ if $i > j$ (entries below diagonal are zero).
- **Strictly Upper Triangular:** $A_{ij} = 0$ if $i \geq j$ (upper triangular with zeros on diagonal).
- **Symmetric:** $A = A^T$ ($A_{ij} = A_{ji}$). Diagonal acts as a mirror.

MULTIPLICATION

Product AB is defined if $\text{columns}(A) = \text{rows}(B)$. Result dim : $\text{rows}(A) \times \text{cols}(B)$.

$$\begin{matrix} A & B & = & C \\ m \times k & k \times n & & m \times n \end{matrix}$$

The entry C_{ij} is the dot product of row i of A and column j of B .

Example:
$$\begin{pmatrix} a & b \\ c & d \\ e & f \end{pmatrix}_{3 \times 2} \begin{pmatrix} g & h & i & m \\ k & l & j & n \end{pmatrix}_{2 \times 4} = \begin{pmatrix} ag + bk & ah + bl & ai + bj & am + bn \\ cg + dk & ch + dl & ci + dj & cm + dn \\ eg + fk & eh + fl & ei + fj & em + fn \end{pmatrix}_{3 \times 4}$$

- If $\mathbf{a}_i$ are **rows** of A , rows of AB are $\mathbf{a}_i B$. $AB = \begin{pmatrix} \mathbf{a}_1 B \\ \vdots \\ \mathbf{a}_m B \end{pmatrix}$
- If $\mathbf{b}_j$ are **columns** of B , columns of AB are $A\mathbf{b}_j$. $AB = (\mathbf{A}\mathbf{b}_1 \ \dots \ \mathbf{A}\mathbf{b}_n)$

Properties: $A^n A^m = A^{n+m}$. BUT generally $(AB)^n \neq A^n B^n$. Matrix multiplication is **not commutative** ($AB \neq BA$).

TRANSPOSE

Definition: $(A^T)_{ij} = A_{ji}$. Rows become columns, columns become rows.

$$A = \begin{pmatrix} 1 & 0 & 3 \\ 4 & -1 & -2 \\ 0 & 0 & 5 \end{pmatrix} \implies A^T = \begin{pmatrix} 1 & 4 & 0 \\ 0 & -1 & 0 \\ 3 & -2 & 5 \end{pmatrix}$$

Properties:

- $(cA)^T = cA^T$
- $(A+B)^T = A^T + B^T$ (Matrices must be the same size)
- $(AB)^T = B^T A^T$ (Reverse order!)

EQUATIONS

Representing linear systems: **Matrix Equation:** $A\mathbf{x} = \mathbf{b}$

$$\text{e.g., } \begin{pmatrix} 3 & 0 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$

Vector Equation: $x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \dots = \mathbf{b}$

$$\text{e.g., } x \begin{pmatrix} 3 \\ 1 \end{pmatrix} + y \begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$

MULTIPLE SYSTEMS

Solving several systems $A\mathbf{x} = \mathbf{b}_i$ with the same A simultaneously. Augment A with all RHS vectors: $[A|\mathbf{b}_1 \ \mathbf{b}_2 \ \dots]$.

$$\text{e.g., For } A = \begin{pmatrix} 1 & 3 & 1 \\ 0 & 4 & -1 \end{pmatrix} \text{ solve for } \mathbf{b}_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \mathbf{b}_2 = \begin{pmatrix} 5 \\ 5 \end{pmatrix}$$

$$\left(\begin{array}{ccc|cc} 1 & 3 & 1 & 0 & 5 \\ 0 & 4 & -1 & 1 & 5 \end{array} \right) \xrightarrow{\text{EROs}} \text{Solve together}$$

INVERSE OF MATRICES

Situation: $AB = I$.

- A is the **left inverse** of B .
- B is the **right inverse** of A .

For non-square matrices, these might exist but are not unique. A **square** matrix A is **invertible** (or **non-singular**) if there exists a unique matrix A^{-1} such that $AA^{-1} = A^{-1}A = I$. If no inverse exists, A is **singular**. Not all square matrices have an inverse.

INVERTIBLE MATRIX MATH

(Assuming A, B invertible, $a \neq 0$ scalar).

- $(A^{-1})^{-1} = A$
- $(aA)^{-1} = \frac{1}{a}A^{-1}$
- $(A^T)^{-1} = (A^{-1})^T$
- $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$ (Reverse order!)

- $(A^n)^{-1} = (A^{-1})^n = A^{-n}$

FINDING INVERSE

2x2 Inverse: If $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $ad - bc \neq 0$, then

$$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

General Method (Gauss-Jordan): For an $n \times n$ matrix A . Form $[A|I]$. Use EROs to transform A into I . The right side becomes A^{-1} .

$$[A|I] \xrightarrow{\text{EROs}} [I|A^{-1}]$$

If A cannot be reduced to I , then A is singular (not invertible). Example (from

notes' result): $A = \begin{pmatrix} 5 & 1 & 0 \\ 2 & 0 & 3 \\ 0 & 0 & -1 \end{pmatrix}$.

$$\left(\begin{array}{ccc|ccc} 5 & 1 & 0 & 1 & 0 & 0 \\ 2 & 0 & 3 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & 0 & 1 \end{array} \right) \xrightarrow{\text{EROs}} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1/2 & 3/2 \\ 0 & 1 & 0 & 1 & -5/2 & -15/2 \\ 0 & 0 & 1 & 0 & 0 & -1 \end{array} \right)$$

$$\text{So, } A^{-1} = \begin{pmatrix} 0 & 1/2 & 3/2 \\ 1 & -5/2 & -15/2 \\ 0 & 0 & -1 \end{pmatrix}.$$

INV. OF ELEMENTARY MATRICES

Elementary matrices perform EROs via multiplication. Their inverses undo the operation.

- E for $R_i \leftarrow R_i + kR_j \implies E^{-1}$ for $R_i \leftarrow R_i - kR_j$.

$$E = \begin{pmatrix} 1 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \implies E^{-1} = \begin{pmatrix} 1 & -3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- E for $R_i \leftrightarrow R_j \implies E^{-1} = E$.

$$E = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \implies E^{-1} = E$$

- E for $R_i \leftarrow cR_i \implies E^{-1}$ for $R_i \leftarrow \frac{1}{c}R_i$.

$$E = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix} \implies E^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1/2 \end{pmatrix}$$

Note: $I^{-1} = I$.

Row Equivalent Matrices

If $A \xrightarrow{\text{EROs}} B$, they are row equivalent. $B = E_k \dots E_2 E_1 A$, and $A = E_1^{-1} E_2^{-1} \dots E_k^{-1} B$. (E_i are elementary matrices).

DETERMINANTS

SHORTCUTS

1x1: $A = (a) \implies \det(A) = a$.

2x2: $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \implies \det(A) = ad - bc$.

3x3 (Sarrus' Rule): Repeat first two columns. Sum products (+) down-right diagonals, subtract products (-) down-left diagonals.

$$A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \implies \begin{vmatrix} a & b & c & a & b \\ d & e & f & d & e \\ g & h & i & g & h \end{vmatrix}$$

$$\det(A) = (aei + bfg + cdh) - (gec + hfa + idb)$$

COFACTOR EXPANSION

General method for any $n \times n$ matrix.

1. Choose any single row or column.

2. For each element a_{ij} in that row/column:

- Minor M_{ij} : Determinant of submatrix removing row i , col j .

- Cofactor $C_{ij} = (-1)^{i+j} M_{ij}$. Signs pattern: $\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$.

3. $\det(A) = \sum a_{ij} C_{ij}$ along the chosen row/column.

Example (Expansion along row 1):

$$\det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = aC_{11} + bC_{12} + cC_{13}$$

$$= a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

$$= a(ei - hf) - b(di - gf) + c(dh - ge)$$

DETERMINANT REDUCTION (Using EROs)

Effect of EROs: Let $A \xrightarrow{\text{ERO}} B$.

- $R_i \leftarrow R_i + kR_j$: $\det(B) = \det(A)$. ($\det(E) = 1$).

- $R_i \leftrightarrow R_j$: $\det(B) = -\det(A)$. ($\det(E) = -1$).

- $R_i \leftarrow cR_i$ ($c \neq 0$): $\det(B) = c \det(A)$. ($\det(E) = c$).

Example: Situation: $A \xrightarrow{E_1: R_1 \leftrightarrow R_3} \xrightarrow{E_2: R_2 \leftarrow R_2 + 5R_1} B$. $B = E_2 E_1 A$. Given $\det(A) = -2$. $\det(E_1) = -1$, $\det(E_2) = 1$. $\det(B) = \det(E_2) \det(E_1) \det(A) = (1) \times (-1) \times (-2) = 2$.

DET MATHS

Let A, B be $n \times n$ matrices, c a scalar.

- $\det(AB) = \det(A) \det(B)$

- $\det(A^{-1}) = \frac{1}{\det(A)} = (\det(A))^{-1}$ (if A invertible)

- $\det(cA) = c^n \det(A)$ (n is dimension of A)

- $\det(A) = \det(A^T)$

- * If A is **triangular**, $\det(A)$ = product of diagonal entries.

3 EUCLIDEAN SPACES

EUCLIDEAN SPACE

$\mathbb{R}^n$ is the space of n -dimensional vectors (usually column vectors $\mathbf{u} = (u_1, \dots, u_n)^T$). Example $A\mathbf{u} = \mathbf{b}$:

$$\begin{pmatrix} 1 & 5 \\ 2 & 10 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \end{pmatrix}$$

RREF of augmented matrix: $\left(\begin{array}{cc|c} 1 & 5 & 3 \\ 0 & 0 & 0 \end{array} \right)$ Solution: $x + 5y = 3$. Let $y = s$ (free),

$s \in \mathbb{R}$. $x = 3 - 5s$. Solution Set: $\left\{ \begin{pmatrix} 3-5s \\ s \end{pmatrix} \mid s \in \mathbb{R} \right\} = \left\{ \begin{pmatrix} 3 \\ 0 \end{pmatrix} + s \begin{pmatrix} -5 \\ 1 \end{pmatrix} \mid s \in \mathbb{R} \right\}$.

$\begin{pmatrix} 3 \\ 0 \end{pmatrix}$ is a **particular solution**. The set is the line through $(3, 0)$ parallel to $(-5, 1)$.

SUBSETS

Describing subsets (lines, planes, etc.) in $\mathbb{R}^n$. **Implicit Form:** Using equations/conditions. e.g., $S = \{(x, y, z) \in \mathbb{R}^3 \mid x = 3, y = z\}$. **Explicit (Parametric)**

Form: Using parameters. e.g., $L = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + s \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \mid s \in \mathbb{R} \right\}$. (Solution set form).

VECTOR MATH

Inner / Dot Product: $\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v} = u_1 v_1 + \dots + u_n v_n$ (Scalar result). Example: $\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} d \\ e \\ f \end{pmatrix} = ad + be + cf$.

Outer Product: $\mathbf{u}\mathbf{v}^T$ ($n \times n$ matrix result). $(\mathbf{u}\mathbf{v}^T)_{ij} = u_i v_j$.

FORMULAS

Norm / Magnitude: $\|\mathbf{u}\| = \sqrt{\mathbf{u} \cdot \mathbf{u}} = \sqrt{u_1^2 + \dots + u_n^2}$.

Unit Vector: Vector with norm 1. $\hat{\mathbf{u}} = \frac{\mathbf{u}}{\|\mathbf{u}\|}$ ($\mathbf{u} \neq \mathbf{0}$).

Distance: $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$.

Angle θ between $\mathbf{u}, \mathbf{v}$: $\theta = \arccos\left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}\right)$.

$\mathbf{u} \cdot \mathbf{v} = 0 \iff \mathbf{u}$ and $\mathbf{v}$ are **orthogonal** ($\theta = 90^\circ$).

ORTHOGONAL (OG) & ORTHONORMAL (ON)

For a set of vectors $\{\mathbf{u}, \mathbf{v}, \mathbf{w}, \dots\}$. **Orthogonal (OG):** Every pair of distinct vectors is orthogonal ($\mathbf{u} \cdot \mathbf{v} = 0$, etc.). **Orthonormal (ON):** Orthogonal AND every vector is a unit vector ($\|\mathbf{u}\| = 1$, etc.).

If a set is orthogonal and contains only non-zero vectors, it is Linearly Independent. If the set is orthonormal, it is Linearly Independent. The set containing the zero vector is never L.I.

NORMALISING

Converting $\mathbf{v} \neq \mathbf{0}$ into a unit vector $\hat{\mathbf{v}}$ in the same direction: $\hat{\mathbf{v}} = \mathbf{v} / \|\mathbf{v}\|$. Example: Set $\{(1, 1, 1), (0, 1, -1), (-2, 1, 1)\}$. Norms $\sqrt{3}, \sqrt{2}, \sqrt{6}$. Normalised (ON) set:

$$\left\{ \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}, \frac{1}{\sqrt{6}} \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} \right\}$$

A set containing $\mathbf{0}$ cannot be normalised.

LINEAR COMBINATION

Linear Combination of $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$: $c_1 \mathbf{v}_1 + \dots + c_k \mathbf{v}_k$ (scalars c_i). **Span(S):** Set of all possible linear combinations of vectors in S . It's a subspace. **Linearly Independent (L.I.):** $c_1 \mathbf{v}_1 + \dots + c_k \mathbf{v}_k = \mathbf{0}$ only has solution $c_1 = \dots = c_k = 0$. (No vector is combo of others). Set must not contain $\mathbf{0}$ (unless it's just $\{\mathbf{0}\}$). Geometric Span (L.I. sets): 1 vector $\rightarrow$ line, 2 vectors $\rightarrow$ plane, 3 vectors $\rightarrow$ volume ($\mathbb{R}^3$). * If $\mathbf{d} = a\mathbf{u} + b\mathbf{v}$, then $\mathbf{d} \in \text{span}\{\mathbf{u}, \mathbf{v}\}$ and coordinates of $\mathbf{d}$ wrt basis $S = \{\mathbf{u}, \mathbf{v}\}$ are $(\mathbf{d})_S = \begin{pmatrix} a \\ b \end{pmatrix}$.

SOLVING SPANS

1. Is $\mathbf{w} \in \text{span}\{\mathbf{u}, \mathbf{v}\}$? Check consistency of $[\mathbf{u} \ \mathbf{v} | \mathbf{w}]$. Yes if consistent.
2. Does $\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = \mathbb{R}^n$? Check if matrix $A = [\mathbf{v}_1 \dots \mathbf{v}_k]$ has a pivot in every row. Requires $k \geq n$. * A set of $< n$ vectors cannot span $\mathbb{R}^n$.
3. Does $\text{span}(U) = \text{span}(V)$? (where $U = \{\mathbf{u}_i\}, V = \{\mathbf{v}_i\}$). Check if every $\mathbf{v}_j \in \text{span}(U)$ AND every $\mathbf{u}_i \in \text{span}(V)$ (check consistency of $[U|V]$ and $[V|U]$). Alternatively, check if basis for $\text{span}(U) = \text{basis for span}(V)$. Find RREF(U^T) and RREF(V^T), compare non-zero rows.

SUBSPACES

Subset $A \subseteq \mathbb{R}^n$ is a subspace if:

1. $\mathbf{0} \in A$.
2. Closed under Addition: $\mathbf{u}, \mathbf{v} \in A \implies \mathbf{u} + \mathbf{v} \in A$.
3. Closed under Scalar Mult: $\mathbf{u} \in A, c \in \mathbb{R} \implies c\mathbf{u} \in A$.

Span of any set of vectors is always a subspace. **Disproving:** Show one condition fails. * Zero space $\{\mathbf{0}\}$ is a subspace. $\text{Span}(\emptyset) = \{\mathbf{0}\}$. * Solution set of $A\mathbf{x} = \mathbf{0}$ (Null Space) is always a subspace.

LINEARLY INDEPENDENT (L.I.)

Recap: $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ is L.I. if $c_1 \mathbf{u}_1 + \dots + c_k \mathbf{u}_k = \mathbf{0}$ only has trivial solution ($c_i = 0$). Check: Matrix $A = [\mathbf{u}_1 \dots \mathbf{u}_k]$. Set is L.I. $\iff A\mathbf{c} = \mathbf{0}$ has only trivial solution $\iff$ RREF of A has pivot in every column $\iff \text{rank}(A) = k$. Notes:

- If $k > n$ in $\mathbb{R}^n$, set is L.D.
- If $\mathbf{0}$ is in the set, it's L.D. (unless it's just $\{\mathbf{0}\}$).

BASIS

Situation: $S = \{\mathbf{v}_1, \dots, \mathbf{v}_k\}$, V is a subspace. S is a **basis** for V if:

1. S spans V ($\text{span}(S) = V$).
2. S is Linearly Independent.

Equivalences: Let $\dim(V) = d$.

- If $S \subseteq V$ has d vectors, S is basis $\iff S$ is L.I. $\iff S$ spans V .
- General check: S L.I. and Spans V .

DIMENSIONS

Dimension $\dim(V)$ = number of vectors in any basis for V . $\dim(\mathbb{R}^n) = n$. * $\dim(\{\mathbf{0}\}) = 0$. Basis is $\emptyset$.

4 VECTOR SPACES

COL & ROW SPACE

Let A be $m \times n$. **Row Space (Row(A)):** Subspace of $\mathbb{R}^n$ spanned by rows of A . $\dim(\text{Row}(A)) = \text{rank}(A)$. **Column Space (Col(A)):** Subspace of $\mathbb{R}^m$ spanned by columns of A . $\dim(\text{Col}(A)) = \text{rank}(A)$. $\text{Col}(A) = \{A\mathbf{u} \mid \mathbf{u} \in \mathbb{R}^n\}$. **Row Equivalence:** If $A \xrightarrow{\text{EROs}} B$: * $\text{Row}(A) = \text{Row}(B)$. * Linear dependence relations among columns are preserved (e.g., if $\mathbf{a}_3 = \mathbf{a}_1 + \mathbf{a}_2$, then $\mathbf{b}_3 = \mathbf{b}_1 + \mathbf{b}_2$). * BUT $\text{Col}(A) \neq \text{Col}(B)$ generally. * System $(A|\mathbf{b})$ consistent $\iff \mathbf{b} \in \text{Col}(A)$. * $\dim(\text{Col}(A)) = \dim(\text{Row}(A)) = \text{rank}(A)$.

BASIS FOR ROW SPACE

1. Find RREF (or REF) of A , call it B . 2. Basis = **non-zero rows** of B .

$$A = \begin{pmatrix} 3 & 1 & 0 & 1 \\ 0 & 2 & 3 & 0 \\ 3 & 5 & 6 & 1 \end{pmatrix} \xrightarrow{\text{ref}} B = \begin{pmatrix} 3 & 1 & 0 & 1 \\ 0 & 2 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Basis for $\text{Row}(A) = \{(3, 1, 0, 1), (0, 2, 3, 0)\}$. Extending basis: Row method useful. Put vectors as rows, add standard basis vectors if needed, find RREF, non-zero rows form basis for combined span.

BASIS FOR COLUMN SPACE

1. Find RREF (or REF) of A , call it B . 2. Identify **pivot columns** in B . 3. Basis = columns from **original matrix A** corresponding to pivot columns of B .

$$A = \begin{pmatrix} 3 & 1 & 0 & 1 \\ 0 & 2 & 3 & 0 \\ 3 & 5 & 6 & 1 \end{pmatrix} \xrightarrow{\text{ref}} B = \begin{pmatrix} \boxed{3} & 1 & 0 & 1 \\ 0 & \boxed{2} & 3 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Pivot columns in B are 1 and 2. Basis for $\text{Col}(A) = \left\{ \begin{pmatrix} 3 \\ 0 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix} \right\}$.

RANK

$\text{rank}(A) = \dim(\text{Col}(A)) = \dim(\text{Row}(A)) = \text{Number of pivots in RREF}(A) = \text{Number of non-zero rows in RREF}(A)$. Properties (A is $m \times n$):

- $\text{rank}(A) \leq \min(m, n)$
- $\text{rank}(A) = \text{rank}(A^T)$
- Only zero matrix has rank 0.

A has **full rank** if $\text{rank}(A) = \min(m, n)$.

FULL RANK

Let A be $m \times n$. Full rank means $\text{rank}(A) = \min(m, n)$. **Case 1: Full Column Rank** ($\text{rank}(A) = n$, requires $m \geq n$)

- RREF looks like $\begin{pmatrix} I_n \\ 0 \end{pmatrix}$ (if $m > n$).
- Columns are L.I.
- $A\mathbf{x} = \mathbf{0}$ only has trivial solution ($\text{Null}(A) = \{\mathbf{0}\}$).
- $A^T A$ is invertible ($n \times n$).
- A has a left inverse.

Case 2: Full Row Rank ($\text{rank}(A) = m$, requires $n \geq m$)

- RREF has pivot in every row. Looks like $[I_m | F]$ after column perm.

- Rows are L.I.
- $\text{Col}(A) = \mathbb{R}^m$ (System $A\mathbf{x} = \mathbf{b}$ is consistent for all $\mathbf{b} \in \mathbb{R}^m$).
- AA^T is invertible ($m \times m$).
- A has a right inverse.

If A is square $n \times n$, full rank means $\text{rank}(A) = n$, RREF is I_n , A is invertible (see Great List). (Note: $\{\mathbf{0}\}$ is the set containing only the zero vector, distinct from empty set $\emptyset$).

NULL SPACE

$\text{Null}(A) = \{\mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} = \mathbf{0}\}$. Subspace of $\mathbb{R}^n$. **Finding Basis for $\text{Null}(A)$:** 1. Solve $A\mathbf{x} = \mathbf{0}$ (find RREF of $[A | \mathbf{0}]$). 2. Write solution in parametric vector form. 3. Basis = vectors associated with free parameters. Example:

$$A = \begin{pmatrix} 5 & 3 & 0 \\ 10 & 6 & 0 \end{pmatrix} \rightarrow \text{RREF} \begin{pmatrix} 1 & 0.6 & 0 \\ 0 & 0 & 0 \end{pmatrix}. x + 0.6y = 0. y = s \text{ (free)}. x = -0.6s.$$

Solution: $\mathbf{x} = s \begin{pmatrix} -0.6 \\ 1 \end{pmatrix}$. Basis = $\left\{ \begin{pmatrix} -0.6 \\ 1 \end{pmatrix} \right\}$. **Nullity**(A) = $\dim(\text{Null}(A)) = \#$ free variables = $\#$ non-pivot columns. * **Rank-Nullity Theorem:** For $m \times n$ matrix A :

$$\text{rank}(A) + \text{Nullity}(A) = n$$

($\#$ pivot cols + $\#$ non-pivot cols = total $\#$ cols).

5 ORTHOGONAL PROJECTION

ORTHO-BASIS

Orthogonal Basis (OGB): Basis that is an orthogonal set. **Orthonormal Basis (ONB):** Basis that is an orthonormal set. Properties:

- Any OG set of non-zero vectors is L.I.
- n non-zero OG vectors in $\mathbb{R}^n$ form an OGB for $\mathbb{R}^n$.
- n ON vectors in $\mathbb{R}^n$ form an ONB for $\mathbb{R}^n$.

SOLVING SHORTCUT (Coordinates w.r.t. Orthogonal Basis)

If $S = \{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ is OGB for W , and $\mathbf{v} \in W$, then $\mathbf{v} = c_1\mathbf{u}_1 + \dots + c_k\mathbf{u}_k$, where coordinates c_k are:

$$c_k = \frac{\mathbf{v} \cdot \mathbf{u}_k}{\mathbf{u}_k \cdot \mathbf{u}_k} = \frac{\mathbf{v} \cdot \mathbf{u}_k}{\|\mathbf{u}_k\|^2}$$

If S is ONB, then $\|\mathbf{u}_k\|^2 = 1$, so: $c_k = \mathbf{v} \cdot \mathbf{u}_k$. Requires $\mathbf{v} \in \text{span}(S)$.

STANDARD BASIS E

$E = \{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ for $\mathbb{R}^n$. $\mathbf{e}_i$ has 1 in i -th position, 0 elsewhere. Example $\mathbb{R}^3$: $E = \{(1, 0, 0)^T, (0, 1, 0)^T, (0, 0, 1)^T\}$. E is an ONB. Coordinates $(\mathbf{v})_E = \mathbf{v}$.

ORTHOGONAL PROJECTION

Decomposition: $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$, where $\hat{\mathbf{y}} \in W$ and $\mathbf{z} \in W^\perp$. $\hat{\mathbf{y}} = \text{proj}_W \mathbf{y}$ is the orthogonal projection of $\mathbf{y}$ onto subspace W . Closest vector in W to $\mathbf{y}$. If $S = \{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ is an **OGB** for W :

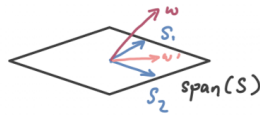
$$\hat{\mathbf{y}} = \text{proj}_W \mathbf{y} = \sum_{i=1}^k \left(\frac{\mathbf{y} \cdot \mathbf{u}_i}{\|\mathbf{u}_i\|^2} \right) \mathbf{u}_i$$

If S is an **ONB** for W :

$$\hat{\mathbf{y}} = \text{proj}_W \mathbf{y} = \sum_{i=1}^k (\mathbf{y} \cdot \mathbf{u}_i) \mathbf{u}_i$$

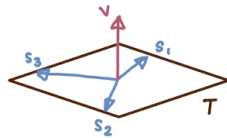
Orthogonal component $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}}$. Example (notes): Project $\mathbf{w} = (1, 1, 1)^T$ onto $S = \text{span}\{\mathbf{s}_1, \mathbf{s}_2\}$ where $\mathbf{s}_1 = (1, 0, 0)^T, \mathbf{s}_2 = (0, 1, 0)^T$. (ONB for xy-plane).

$$\mathbf{w}' = (\mathbf{w} \cdot \mathbf{s}_1)\mathbf{s}_1 + (\mathbf{w} \cdot \mathbf{s}_2)\mathbf{s}_2 = (1)\mathbf{s}_1 + (1)\mathbf{s}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}.$$



OG SUBSPACE (Orthogonal Complement $W^\perp$)

$W^\perp = \{\mathbf{v} \in \mathbb{R}^n \mid \mathbf{v} \cdot \mathbf{w} = 0 \text{ for all } \mathbf{w} \in W\}$. $W^\perp$ is a subspace. **Orthogonal Decomposition:** $\mathbf{v} = \hat{\mathbf{v}} + \mathbf{z}$ where $\hat{\mathbf{v}} = \text{proj}_W \mathbf{v} \in W$ and $\mathbf{z} = \text{proj}_{W^\perp} \mathbf{v} \in W^\perp$. $\dim(W) + \dim(W^\perp) = n$.



Fundamental Subspaces Connection: For any $m \times n$ matrix A :

- $(\text{Row}(A))^\perp = \text{Null}(A)$
- $(\text{Col}(A))^\perp = \text{Null}(A^T)$

Example: If $V = \text{Col}(A)$, then $\mathbf{w} \perp V \iff \mathbf{w} \in \text{Null}(A^T)$ (i.e., $A^T \mathbf{w} = \mathbf{0}$).

GRAM-SCHMIDT PROCESS

Converts a basis $S = \{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ for W into an OGB $V = \{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ for W .

1. $\mathbf{v}_1 = \mathbf{u}_1$.
2. $\mathbf{v}_2 = \mathbf{u}_2 - \text{proj}_{\mathbf{v}_1} \mathbf{u}_2 = \mathbf{u}_2 - \frac{\mathbf{u}_2 \cdot \mathbf{v}_1}{\|\mathbf{v}_1\|^2} \mathbf{v}_1$.
3. $\mathbf{v}_3 = \mathbf{u}_3 - \text{proj}_{\mathbf{v}_1} \mathbf{u}_3 - \text{proj}_{\mathbf{v}_2} \mathbf{u}_3 = \mathbf{u}_3 - \frac{\mathbf{u}_3 \cdot \mathbf{v}_1}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 - \frac{\mathbf{u}_3 \cdot \mathbf{v}_2}{\|\mathbf{v}_2\|^2} \mathbf{v}_2$.

4. Continue: $\mathbf{v}_j = \mathbf{u}_j - \sum_{i=1}^{j-1} \frac{\mathbf{u}_j \cdot \mathbf{v}_i}{\|\mathbf{v}_i\|^2} \mathbf{v}_i$.

To get ONB, normalize each $\mathbf{v}_i$: $\hat{\mathbf{v}}_i = \mathbf{v}_i / \|\mathbf{v}_i\|$.

LEAST SQUARES SOLUTIONS

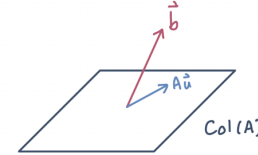
For inconsistent system $A\mathbf{x} = \mathbf{b}$, find best approximate solution $\hat{\mathbf{x}}$ minimizing $\|A\mathbf{x} - \mathbf{b}\|$. $A\hat{\mathbf{x}} = \hat{\mathbf{b}} = \text{proj}_{\text{Col}(A)} \mathbf{b}$. $\hat{\mathbf{x}}$ is the least squares solution. Solve the **Normal Equations** (always consistent):

$$A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$$

If columns of A are L.I., $A^T A$ is invertible, unique solution:

$$\hat{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b}$$

If columns L.D., infinitely many $\hat{\mathbf{x}}$, but $\hat{\mathbf{b}} = A\hat{\mathbf{x}}$ is unique. **MATLAB:** $\hat{\mathbf{x}} = A \setminus \mathbf{b}$ or $\text{pinv}(A) * \mathbf{b}$.



Example (notes): $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$. $A^T A = I_2$. $A^T \mathbf{b} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$. Normal Eq:

$$I_2 \hat{\mathbf{x}} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \implies \hat{\mathbf{x}} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}. \text{ Projection: } \hat{\mathbf{b}} = A\hat{\mathbf{x}} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}.$$

6 DIAGONALIZATION

EIGENVALUES & VECTORS

Situation: $A\mathbf{v} = \lambda\mathbf{v}$, for $n \times n$ matrix A . Conditions: λ is a scalar (eigenvalue), $\mathbf{v} \neq \mathbf{0}$ is a vector (eigenvector). Eigenvector $\mathbf{v}$ is scaled by λ under transformation A . * Eigenvalues of a triangular matrix are its diagonal entries.

CHARACTERISTIC POLYNOMIAL

$A\mathbf{v} = \lambda\mathbf{v} \iff (A - \lambda I)\mathbf{v} = \mathbf{0}$. Has non-trivial solution $\mathbf{v}$ iff $(A - \lambda I)$ is singular. $\iff \det(A - \lambda I) = 0$. **Characteristic Polynomial:** $p(\lambda) = \det(A - \lambda I)$. (Degree n). **Eigenvalues** are the roots of $p(\lambda) = 0$. Example: $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

$p(\lambda) = \det \begin{pmatrix} 1 - \lambda & 1 \\ 0 & 1 - \lambda \end{pmatrix} = (1 - \lambda)^2$. Eigenvalue $\lambda = 1$ (repeated). * λ is eigenvalue $\iff (A - \lambda I)$ is singular / non-invertible.

ALGEBRAIC MULTIPLICITY (r_λ)

r_λ = multiplicity of λ as a root of $p(\lambda) = 0$. * Sum of all r_λ for distinct eigenvalues = n . ($\sum r_\lambda = n$).

EIGENSPACE (E_λ)

Set of all eigenvectors for λ , plus **0**. $E_\lambda = \{\mathbf{v} \mid A\mathbf{v} = \lambda\mathbf{v}\} = \text{Null}(A - \lambda I)$. Subspace. Basis for E_λ : Find basis for $\text{Null}(A - \lambda I)$. **Geometric Multiplicity** (g_λ): $g_\lambda = \dim(E_\lambda) = \text{Nullity}(A - \lambda I)$. $g_\lambda = \#$ L.I. eigenvectors for λ . Inequality:

$1 \leq g_\lambda \leq r_\lambda$. Example: $A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$. $\lambda_1 = 1$ ($r_1 = 2$), $\lambda_2 = 2$ ($r_2 = 1$).

$E_1 = \text{Null}(A - I) = \text{span}\{(1, 0, 0)^T\}$. $g_1 = 1$. $E_2 = \text{Null}(A - 2I) = \text{span}\{(0, 0, 1)^T\}$. $g_2 = 1$. (Note: $g_1 = 1 < r_1 = 2$, so this A is not diagonalizable).

DIAGONALIZATION

$n \times n$ matrix A is **diagonalizable** if $A = PDP^{-1}$ for some invertible P and diagonal D . $\iff A$ has n linearly independent eigenvectors. $\iff g_\lambda = r_\lambda$ for all eigenvalues λ . If diagonalizable:

- Columns of P are n L.I. eigenvectors. $P = [\mathbf{p}_1 \dots \mathbf{p}_n]$.
- Diagonal entries of D are corresponding eigenvalues. $D = \text{diag}(\lambda_1, \dots, \lambda_n)$.
- $A\mathbf{p}_i = \lambda_i\mathbf{p}_i$.

* O, I, D are always diagonalizable.

CHECKING DIAGONALIZABILITY

Check if total number of L.I. eigenvectors (sum of g_λ) equals n . Or, check if $g_\lambda = r_\lambda$ for all λ . If A has n distinct eigenvalues, it's always diagonalizable ($r_\lambda = 1 \implies g_\lambda = 1$).

FINDING P AND D

1. Find eigenvalues λ and their algebraic multiplicities r_λ . 2. Find basis for each eigenspace E_λ . Get geometric multiplicity $g_\lambda = \dim(E_\lambda)$. 3. Check if $\sum g_\lambda = n$ (or $g_\lambda = r_\lambda$ for all λ). If not, stop (not diagonalizable). 4. If yes, construct P with basis eigenvectors as columns. 5. Construct D with corresponding eigenvalues on diagonal (same order as P). Example Structure (based on notes' implied example): A leads to $\lambda = 0$ ($r = 1, g = 1$), $\lambda = 2$ ($r = 2, g = 2$). $E_0 = \text{span}\{\mathbf{p}_1 = (1, 1, 0)^T\}$. $E_2 = \text{span}\{\mathbf{p}_2 = (1, 0, 0)^T, \mathbf{p}_3 = (0, 1, 1)^T\}$. Diagonalizable. $P = [\mathbf{p}_1 \ \mathbf{p}_2 \ \mathbf{p}_3] = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}$. $D = \text{diag}(0, 2, 2) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$.

ORTHOGONAL DIAGONALIZATION

Orthogonal Matrix (OGM): Square matrix P where columns form an ONB. Properties: $P^T P = I \iff P^{-1} = P^T$. Rows also form ONB. **Orthogonally Diagonalizable:** $A = PDP^T$ where P is orthogonal, D diagonal. **Spectral Theorem:** A is orthogonally diagonalizable $\iff A$ is **symmetric** ($A = A^T$). For symmetric A : Eigenvalues are real. Eigenvectors for distinct λ are orthogonal.

PROCESS (Orthogonal Diagonalization)

For symmetric A : 1. Find eigenvalues λ . 2. Find basis for each eigenspace E_λ . 3. Within each E_λ , use Gram-Schmidt to get an OGB. 4. Normalize all OGB vectors to get an ONB of eigenvectors for $\mathbb{R}^n$. 5. P = matrix with these ONB vectors as columns. (P will be orthogonal). 6. D = diagonal matrix of corresponding eigenvalues. Check $A = PDP^T$.

APPLICATION OF DIAG

Powers of A: $A = PDP^{-1} \implies A^k = PD^k P^{-1}$. $D^k = \text{diag}(\lambda_i^k)$. **Linear Recurrence:** $\mathbf{x}_n = A\mathbf{x}_{n-1} \implies \mathbf{x}_n = A^n \mathbf{x}_0 = PD^n P^{-1} \mathbf{x}_0$. **Markov Chains:** State vector $\mathbf{x}_k$. $\mathbf{x}_{k+1} = A\mathbf{x}_k$ (A is transition matrix). $\mathbf{x}_k = A^k \mathbf{x}_0$. Diagonalization helps find $\lim_{k \rightarrow \infty} \mathbf{x}_k$ (steady state).

7 LINEAR DIFFERENTIAL EQUATIONS

EQN TO MATX

System $\mathbf{x}' = A\mathbf{x}$. e.g., $x' = x + y, y' = 2x - y \implies \begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$.

EIGEN-WAY SOLUTION (Distinct Real Eigenvalues)

1. Find eigenvalues λ_i and eigenvectors $\mathbf{v}_i$ of A . 2. Basic solutions: $\mathbf{x}_i(t) = \mathbf{v}_i e^{\lambda_i t}$. 3. General Solution: $\mathbf{x}(t) = c_1 \mathbf{v}_1 e^{\lambda_1 t} + \dots + c_n \mathbf{v}_n e^{\lambda_n t}$. Example (notes): $A = \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix}$. $\lambda_1 = 1, \mathbf{v}_1 = (1, -1)^T$. $\lambda_2 = 2, \mathbf{v}_2 = (0, 1)^T$. General Sol: $\mathbf{x}(t) = c_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^t + c_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{2t}$.

MOTHER OF SOLUTIONS (Fundamental Set)

F.S.S.: Set of n L.I. solutions $\{\mathbf{x}_1(t), \dots, \mathbf{x}_n(t)\}$. General Solution: $\mathbf{x}(t) = \sum c_i \mathbf{x}_i(t)$. **Initial Conditions:** $\mathbf{x}(t_0) = \mathbf{b}$ determines unique c_i .

CHECK SET IS L.I. (Wronskian)

For solutions $\{\mathbf{x}_1(t), \dots, \mathbf{x}_n(t)\}$ of $\mathbf{x}' = A\mathbf{x}$. Form matrix $X(t) = [\mathbf{x}_1(t) \dots \mathbf{x}_n(t)]$. **Wronskian** $W(t) = \det(X(t))$. Set is L.I. $\iff W(t) \neq 0$ for some/any t . Example (notes, constant vectors): Check L.I. of $\{a\mathbf{y}_1, b\mathbf{y}_2, c\mathbf{y}_3\}$, where $\mathbf{y}_1 = (1, 0, 1)^T, \mathbf{y}_2 = (0, 1, 1)^T, \mathbf{y}_3 = (1, 1, 0)^T$. $X = \begin{pmatrix} a & 0 & c \\ 0 & b & c \\ a & b & 0 \end{pmatrix}$. $\det(X) = -2abc$. L.I. if $a, b, c \neq 0$.

Complex Eigenvalues

Situation: $\lambda = a \pm bi$ ($b \neq 0$). Eigenvector $\mathbf{v} = \mathbf{v}_r + i\mathbf{v}_i$. Pair $\lambda, \bar{\lambda}$ gives two real L.I. solutions:

$$\mathbf{x}_1(t) = e^{at}(\cos(bt)\mathbf{v}_r - \sin(bt)\mathbf{v}_i)$$

$$\mathbf{x}_2(t) = e^{at}(\sin(bt)\mathbf{v}_r + \cos(bt)\mathbf{v}_i)$$

Use these in the general solution.

Repeated Eigenvalues

Situation: $r_\lambda > 1$. If $g_\lambda = r_\lambda$: Find r_λ L.I. eigenvectors $\mathbf{v}_j$, get r_λ solutions $\mathbf{v}_j e^{\lambda t}$. If $g_\lambda < r_\lambda$ (defective): Need generalized eigenvectors. Case $r_\lambda = 2, g_\lambda = 1$: 1. Find eigenvector $\mathbf{v}_1$ for λ . Sol 1: $\mathbf{x}_1(t) = \mathbf{v}_1 e^{\lambda t}$. 2. Find generalized eigenvector $\mathbf{v}_2$ by solving $(A - \lambda I)\mathbf{v}_2 = \mathbf{v}_1$. 3. Sol 2: $\mathbf{x}_2(t) = (\mathbf{v}_1 t + \mathbf{v}_2) e^{\lambda t}$. General solution involves $c_1 \mathbf{x}_1 + c_2 \mathbf{x}_2$. Example (notes): $A = \begin{pmatrix} -1 & 1 \\ -4 & 3 \end{pmatrix}$. $\lambda = 1$ ($r = 2$).

$$E_1 = \text{span}\{\mathbf{v}_1 = (1, 2)^T\}. \quad g_1 = 1. \quad \text{Solve } (A - I)\mathbf{v}_2 = \mathbf{v}_1 \implies \begin{pmatrix} -2 & 1 \\ -4 & 2 \end{pmatrix} \mathbf{v}_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

Solution $\mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ (particular solution, e.g., setting free var=0). Solutions:

$$\mathbf{x}_1(t) = \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^t. \quad \mathbf{x}_2(t) = \left(\begin{pmatrix} 1 \\ 2 \end{pmatrix} t + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) e^t.$$

Found: <https://github.com/brianstm/NUS.git>